A Novel Hybrid Approach for Non-Destructive Gold Purity Assessment Using Stereovision and Ultrasonic Analysis

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Abstract

Conventional fire assay techniques, although accurate, are sample-destructive and involve dangerous chemical procedures, thus being ineffective for precious articles and large-scale commercial analysis. In the foregoing presentation, an original non-destructive testing approach has been developed for evaluating the gold purity and genuineness of articles based on a combination of stereoscopic three-dimensional reconstructions and ultrasonic spectroscopy. In the developed approach, a two-fold validation sub-system has been developed involving two main analysis parts: a stereovision analysis module implemented on the basis of two-webcam imaging of defined resolution along with Scale-Invariant Feature Transforms to calculate the exact volumetric density, and an ultrasonic analysis part involving Fast Fourier Transforms of the signal developed by the use of piezoelectric transducers to detect the internal acoustic characteristics of pure gold or specific gold alloys. A new hybrid data combination approach has also been developed involving a combination of measures of geometric compaction along with frequency components to define a two-level testing procedure to distinguish between pure gold, alloy commodities, and electroplated articles.

It showcases scalability ranging from small artisanal workshops to hallmarking units, with immense opportunities for automation of prediction accuracy for gold purity levels via machine learning implementation. Its non-destructive manner shares numerous benefits such as no sample destruction, lower operational expenditure, improved fraud detection, improved usability for SMEs, and compatibility with current environmental, health, and safety regulations due to the elimination of chemical reagents. It has immense applications for providing an alternative solution for destructive testing of gold for purposes of jewellery authentication, hallmark testing, and artifact conservation.

Keywords: Non-destructive testing, gold assay, stereovision, ultrasonic spectroscopy, SIFT algorithm, jewellery authentication, alloy discrimination, fusion of heterogeneous data sources